

PneumoScan-AI — DenseNet121 Pneumonia Detection Model Report

This document summarizes the development, training, evaluation, and explainability analysis of the PneumoScan-AI DenseNet121 deep learning model for pneumonia detection using chest X-ray images.

1. Project Overview

PneumoScan-AI is an AI-powered medical imaging system developed using DenseNet121 transfer learning architecture for automated pneumonia detection from chest X-ray images. The project integrates explainable AI using Grad-CAM visualizations to improve model interpretability for healthcare applications.

2. Technologies Used

- TensorFlow / Keras
- DenseNet121 Transfer Learning
- Python
- Google Colab GPU (T4)
- Grad-CAM Explainability
- NumPy, Matplotlib, Scikit-learn

3. Model Architecture

The model uses DenseNet121 pretrained on ImageNet as the feature extraction backbone. Transfer learning was applied by initially freezing the backbone layers and training a custom classification head. Fine-tuning was later performed on upper DenseNet layers using a smaller learning rate to adapt the model to chest X-ray features.

4. Training Strategy

Phase 1: Frozen training of classifier head.

Phase 2: Fine-tuning top DenseNet layers.

- Loss Function: Binary Crossentropy
- Optimizer: Adam
- Evaluation Metrics: Accuracy, AUC, Precision, Recall

5. Model Evaluation Results

Final evaluation results obtained on the test dataset are summarized below:

- Accuracy : 91.83%
- AUC-ROC : 96.24%
- Precision : 92.48%
- Recall : 94.62%
- Loss : 0.2434

6. Explainable AI (Grad-CAM)

Grad-CAM heatmaps were generated to visualize which lung regions influenced the model's prediction decisions. This improves transparency and trustworthiness of the AI system for healthcare-related use cases.

7. Conclusion

The DenseNet121 based PnemoScan-AI system achieved strong pneumonia classification performance with high AUC and recall values. The integration of Grad-CAM explainability enhances the interpretability of predictions, making the system more suitable for real world AI-assisted healthcare applications.

8. Screenshots and Visual Evidence

